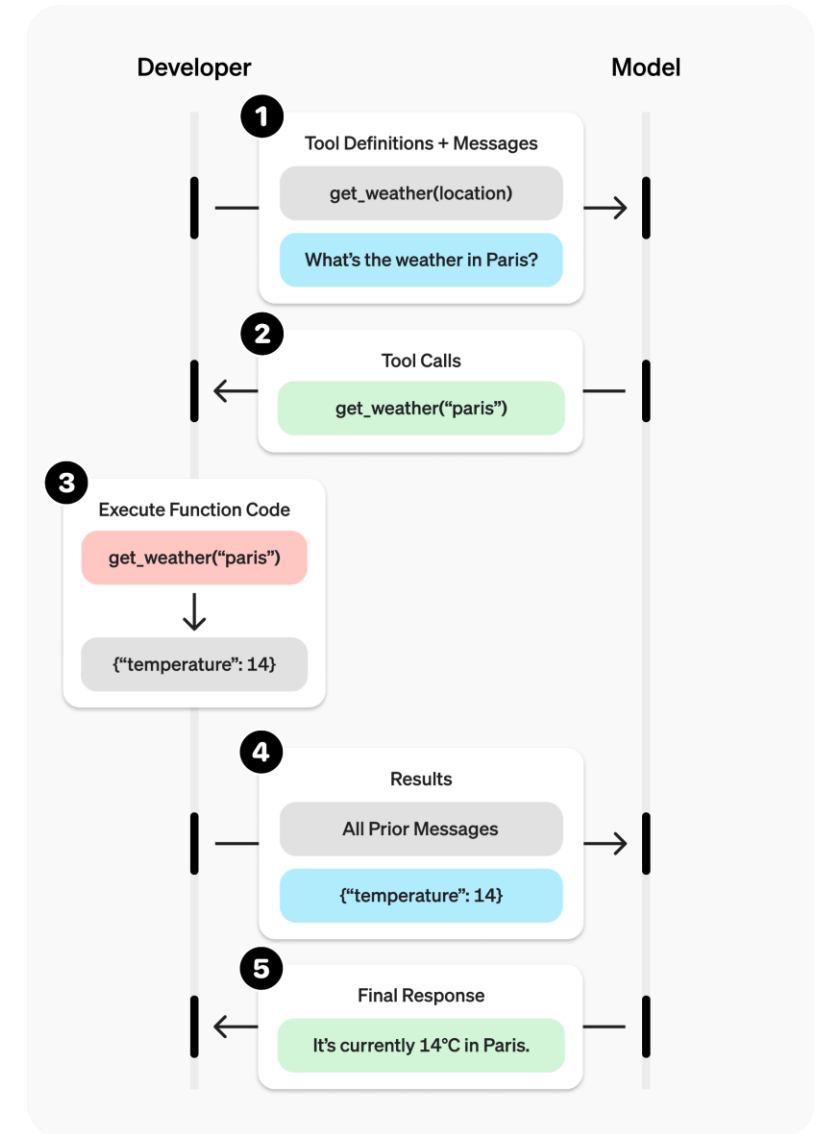
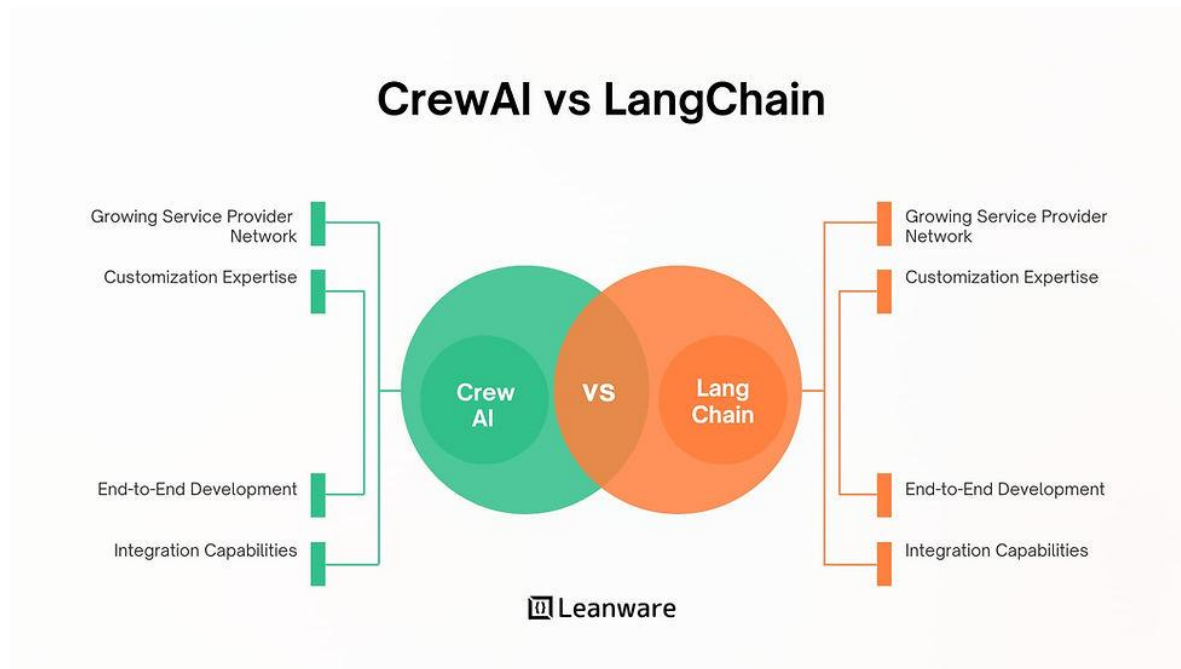


Tool Use & Function Calling: The AI's Hands

- LLMs are naturally "stuck" in their training data; Tool Use allows them to interact with the real world in real-time.
- **Function Calling:** The model detects when it needs a tool and generates structured data (like JSON) to call a function (e.g., a Calculator or a Database).
- This allows the AI to perform specific actions like searching the web, executing code, or checking a user's calendar.

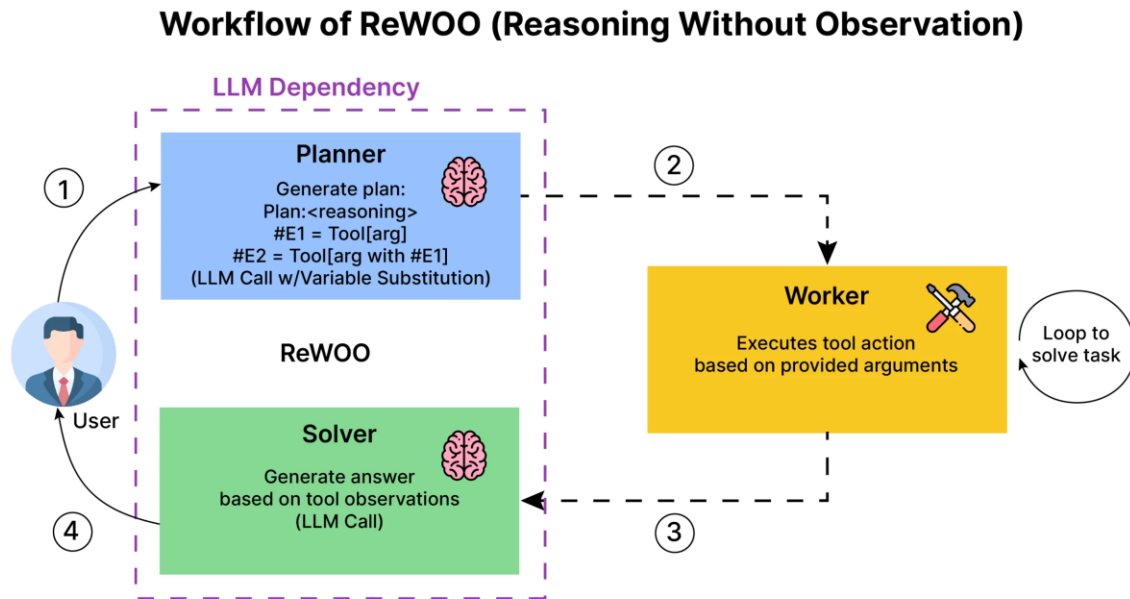


Multi-Agent Systems: CrewAI and LangChain



- Instead of asking one AI to do everything, we assign specific **Roles** (e.g., Researcher, Writer, Quality Analyst).
- Frameworks like **LangChain** and **CrewAI** manage the "handoffs" and communication between these agents.
- This collaborative approach reduces errors and allows for complex, multi-stage workflows that mimic a human team.

Planning & Memory: Achieving Long-Term Goals



- **Planning**: The agent breaks a high-level goal into smaller, actionable tasks (Task Decomposition).
- **ReAct Pattern**: The model follows a "Reason + Act" loop—it thinks about what to do, takes an action, observes the result, and iterates.
- **Memory**: Agents use short-term memory (context window) and long-term memory (Vector DBs) to remember what they have already accomplished.